Lecture 7 Exercise Solutions

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1 Important Equations

Hypothesis Test Case - Known Variance

The test statistic is given as:

$$Z_0 = \frac{\hat{\mu} - \mu}{\sigma/\sqrt{n}} = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1) \tag{1}$$

Where μ_0 is the value for the mean you want to test and s is the unknown population standard deviation.

Hypothesis Test Case - Unknown Variance

The test statistic is given as:

$$T_0 = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} \sim t_{n-1} \tag{2}$$

This is distributed as a student-t distribution.Where μ_0 is the value for the mean you want to test and σ is the known population standard deviation. The variance is estimated as:

$$s^2 = \hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2 \sim \frac{\sigma^2}{\sqrt{n}}\chi_{n-1} \tag{3}$$

Hypothesis Test for the Variance

We know that the sample variance is distributed as:

$$s^2 \sim \frac{\sigma^2}{\sqrt{n}}\chi_{n-1}$$

We can make a hypothesis test on the variance with:

$$H_0 : \sigma^2 = \sigma_0^2, \qquad H_1 \, \sigma^2 \neq \sigma_0^2$$

With the following test statistic:

$$X_0^2 = (n-1)\frac{s^2}{\sigma^2} = (n-1)\frac{s^2}{\sigma_0^2} \sim \chi_{n-1} \tag{4}$$

Hypothesis Test on Population Proportion

For a random variable following the bionomial distribution it is possible to make a hypothesis test on the population proportion p. This could for example be occurence of defectives. First we set up a hypothesis:

$$H_0 : \ p = p_0, \qquad H_1 \ p \neq p_0$$

The test statistic is based on a normal approximation of the binomial:

$$Z_0 = \frac{X - np_0}{\sqrt{np_0(1-p_0)}} \tag{5}$$

This approximation will be valid as long as p is not very close to 0 or 1. Here X is the number of samples in a random sample of size n that belongs to the class of p.

2 Exercise 9.2.2

A hypothesis will be used to test that a population mean equals 5 against the alternative that the population mean is less than 5 with known variance σ^2. What is the critical value for the test statistic Z_0 for the following significance level?

a. 0.01

This is a lower one-sided case with hypothesis and alternative hypothesis:

$$H_0 : \ \mu = 5, \qquad H_1 : \ \mu < 5$$

Since the variance is known, the test statistic Z_0 follows the normal distribution. We can therefore solve the problem using the standard normal distribution to find the critical value for the test statistic in MATLAB. Because it is a lower one-sided test we find it as:

$$Z_{0.01} = \text{norminv}(\alpha) = \text{norminv}(0.01) = -2.3263$$

We do not divide α by two like in the two sided case. This is because in the one-sided case all the area is in one of the tails.

b. 0.05

We use the same command in MATLAB:

$$Z_{0.05} = \text{norminv}(0.05) = -1.6449$$

c. 0.10

$$Z_{0.1} = \text{norminv}(0.1) = -1.2816$$

3 Exercise 9.2.3

Output from a software package

Test of $\mu = 20$ vs. $\mu > 20$

Variable	N	Mean	StDev	SE mean	Z	P
x	10	19.889	?	0.237	?	?

a. Fill in the missing items. What conclusions would you draw?

From the central limit theorem (CLT) we know that for large number of samples n the mean is approximately normal distributed as:

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Meaning we can find our standard deviation from our SE mean (standard error mean or standard deviation of the mean):

$$\text{SE mean} = \frac{s}{\sqrt{N}} \Rightarrow s = \text{SE mean} \cdot \sqrt{N} = 0.237 \cdot \sqrt{10} = 0.75$$

The test statistic is calculated using eq. 2:

$$t_0 = \frac{19.889 - 20}{0.75/\sqrt{10}} = -0.47$$

Because this is a upper-tailed test we find the p-value as (see page 209 in the book):

$$P = 1 - \Phi(z_0) = 1 - \text{tcdf}(-0.47, 10 - 1) = 0.6752$$

The table is then:

Variable	N	Mean	StDev	SE mean	Z	P
x	10	19.889	0.75	0.237	-0.47	0.6808

From this because the p-value is so high we cannot reject H_0. The p-value can be compared informally with a significance value such as for example $\alpha = 0.05$. From this we see the p-value is very high.

b. Is this a one-sided or a two-sided test?

Because we have the following hypothesis and alternative hypothesis from before:

$$H_0 : \ \mu = 20, \qquad H_1 : \ \mu > 20$$

It is a one-sided test.

c. Construct a 95% two-sided CI on the mean.

We first calculate in MATLAB with:

$$t_{n-1,\frac{\alpha}{2}} = \text{tinv}\left(1 - \frac{\alpha}{2}, n-1\right) = \text{tinv}\left(1 - \frac{0.05}{2}, 10-1\right) = 2.2622$$

Where we use student-t distribution since the variance is estimated. Now the two-sided CI on the mean can be calculated as:

$$\left[\bar{X} \pm t_{n-1,\frac{\alpha}{2}} \frac{s}{\sqrt{n}}\right] = \left[19.889 \pm 2.2622 \cdot \frac{0.75}{\sqrt{10}}\right] = [19.35; 20.43]$$

d. What would the p-value be if the alternative hypothesis is $H_1 : \mu \neq 20$?

For a two-sided test the p-value is given as (see page 209 in book):

$$P = 2\left[1 - \text{tcdf}(|t_0|, n-1)\right] = 2\left[1 - \text{tcdf}(|-0.47|, 10-1)\right] = 0.6495$$

4 Exercise 9.2.4

For the hypothesis test $H_0 : \mu = 10$ against $H_1 : \mu > 10$ and variance known, calculate the P-value for each of the following test statistics.

a. $z_0 = 2.05$**.**

This is an upper-tailed case meaning the P-value is (see page 209 in book):

$$P = 1 - \text{normcdf}(z_0) = 1 - \text{normcdf}(2.05) = 0.0202$$

b. $z_0 = -1.84$**.**

$$P = 1 - \text{normcdf}(z_0) = 1 - \text{normcdf}(-1.84) = 0.9671$$

c. $z_0 = 0.4$**.**

$$P = 1 - \text{normcdf}(z_0) = 1 - \text{normcdf}(0.4) = 0.3446$$

5 Exercise 9.2.6

A random sample of $n = 15$ **is tested and** $\bar{x} = 2.78$**. It is known that** $\sigma = 0.9$ **and that** X **is normally distributed.**

a. Test $H_0 : \mu = 3$ **versus** $h_1 : \mu \neq 3$ **using** $\alpha = 0.05$**.**

This is a case with known variance meaning we calculate the test statistic using eq. 1:

$$z_0 = \frac{2.78 - 3}{0.9/\sqrt{15}} = -0.9467$$

We now with MATLAB calculate:

$$z_{\frac{\alpha}{2}} = \text{norminv}\left(1 - \frac{\alpha}{2}\right) = \text{norminv}\left(1 - \frac{0.05}{2}\right) = 1.96$$

Because $|z_0| \leq z_{\frac{\alpha}{2}}$ then we have to accept H_0.

6 Exercise 9.2.7

The life in hours of a battery is known to be approximately normal distributed with std. dev. $\sigma = 1.25$ **hours. A random sample of 10 batteries has a mean life of** $\bar{x} = 40.5$ **hours.**

a. Is there evidence to support the claim that battery life exceeds 40 hours. Use $\alpha = 0.05$.

We have the null hypothesis:

$$H_0 : \ \mu \geq 40 \, \text{hours}$$

Since the variance is known through the std. dev. we can use eq. 1 for the test statistic:

$$z_0 = \frac{40.5 - 40}{1.25/\sqrt{10}} = 1.2649$$

With MATLAB we calculate for an upper-tailed test with $\alpha = 0.05$:

$$z_\alpha = \text{norminv}\,(1 - \alpha) = \text{norminv}\,(1 - 0.05) = 1.6449$$

Since $|z_0| < z_{\frac{\alpha}{2}}$ then there is not evidence to support the claim that the battery life exceeds 40 hours.

b. What is the P-value for the test in part a?

From page 209 in the book the P-value for an upper-tailed test is given using MATLAB as:

$$P = 1 - \text{normcdf}\,(z_0) = 1 - \text{normcdf}\,(1.2649) = 0.103$$

c. What is the β-error for the test in part a if the true mean is 42 hours?

We can calculate the probability of failing to reject H_0 when it is false i.e. H_1 is true. This error is also called the type II error. This means the test statistic should take a value in the acceptance region. For this upper-tailed case with $\delta = 40\,\text{hours} - 42\,\text{hours} = -2\,\text{hours}$ then the probability is:

$$\begin{aligned}\beta = P(H_0|H_1) &= P\left(Z_0 < z_\alpha + \frac{\delta}{\sigma/\sqrt{n}}\right) = \text{normcdf}\left(z_\alpha + \frac{\delta}{\sigma/\sqrt{n}}\right) \\ &= \text{normcdf}\left(1.6449 + \frac{-2}{1.25/\sqrt{10}}\right) = 3.2 \cdot 10^{-4}\end{aligned}$$

d. What sample size would be required to ensure that β does not exceed

0.10 if the true mean is 44 hours?

From eq. 9.23 on page 213 in the book the required sample size is:

$$n \approx \frac{(z_\alpha + z_\beta)^2 \sigma^2}{\delta^2}$$

Where because β should not exceed 0.10:

$$z_\beta = \text{norminv}(1 - 0.1) = 1.2816$$

That means with $\delta = 40 - 44 = -4$:

$$n \approx \frac{(1.6449 + 1.2816)^2 \cdot 1.25^2}{(-4)^2} = 0.8364$$

Which when rounded up is since sample size should be integer $n \approx 1$.

e. Explain how you could answer part a with a confidence bound on battery life.

Because the variance is known the ons-sided CI is:

$$[\bar{x} - z_\alpha \frac{\sigma}{\sqrt{n}}; \infty] = [40.5 - 1.6449 \cdot \frac{1.25}{\sqrt{10}}; \infty] = [39.85; \infty]$$

7 Exercise 9.3.1

A hypothesis will be used to test that a population mean equals 10 against the alternative that the population mean is greater than 10 with unknown variance. What is the critical value for the test statistic T_0 for the following significance levels?

a. $\alpha = 0.01$ **and** $n = 20$

This is a one-sided case and because the variance is unknown we use the student-t distribution. I calculate using MATLAB which needs the significance level and degrees of freedom $df = n - 1$:

$$t_0 = \text{tinv}(1 - \alpha, n - 1) = \text{tinv}(1 - 0.01, 20 - 1) = 2.5395$$

b. $\alpha = 0.05$ **and** $n = 12$

$$t_0 = \text{tinv}(1 - \alpha, n - 1) = \text{tinv}(1 - 0.05, 12 - 1) = 1.7959$$

c. $\alpha = 0.10$ **and** $n = 15$

$$t_0 = \text{tinv}(1-\alpha, n-1) = \text{tinv}(1-0.10, 15-1) = 1.345$$

8 Exercise 9.3.7

Look at the data provided for 9.3.7 on itslearning (ch09.xlsx).

a. Test the hypothesis that the mean is 300. Use $\alpha = 0.05$**.**

Data in MATLAB format with $n = 27$ samples:

x = [421 452.6 456.1 494.6 373.8 447.8 687.6 705.7 879 88.8 90.5 110.7 96.4 81.7 102.4 296 273 268 227.5 279.3 241 296 317 290.9 256.5 258.5 296];

First we set up the hypothesis:

$$H_0: \ \mu = 300, \qquad H_1: \ \mu \neq 300$$

Now we find the sample mean in MATLAB as:

$$\bar{x} = \text{mean}(x) = 325.4963$$

Now we estimate the sample standard deviation:

$$s = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2} = 198.7855$$

The test statistic is calculated for the case of unknown variance using eq. 2:

$$t_0 = \frac{325.5963 - 300}{198.7855/\sqrt{27}} = 0.6665$$

We can now calculate the critical value using MATLAB:

$$t_{n-1,\frac{\alpha}{2}} = \text{tinv}\left(1 - \frac{\alpha}{2}, n-1\right) = \text{tinv}\left(1 - \frac{0.05}{2}, 27-1\right) = 2.0555$$

Because $|t_0| < t_{n-1,\frac{\alpha}{2}}$ we have to accept H_0.

b. Which is the smallest level of significance at which you would be willing to reject the null hypothesis?

We can informally compare α with a P-value. The P-value is calculated as:

$$P = 2\left[1 - \text{tcdf}(t_0, n-1)\right] = 2\left[1 - \text{tcdf}(0.6665, 27-1)\right] = 0.511$$

This means the smallest level of significance where we would be willing to reject the null hypothesis is 0.511.

c. Explain how to answer (a) with a CI.

A $100(1-\alpha)\%$ CI gives a range of possible values for the true mean based on the sample data. It can be calculated as:

$$\left[\bar{x} \pm t_0 \frac{s}{\sqrt{n}}\right] = \left[325.4963 \pm 0.6665 \frac{198.7855}{\sqrt{27}}\right] = [246.86; 404.13]$$

9 Exercise 9.4.1

Consider the test $H_0 : \sigma^2 = 7$ against $H_1 : \sigma^2 \neq 7$. What are the critical values for the test statistic χ_0^2 for the following significance levels?

a. $\alpha = 0.01$ and $n = 20$.

This exercise is solved using MATLAB's "chi2inv" command which requires as input a significance level and degrees of freedom. This is a two sided test meaning there is an upper and lower critical value. First the upper critical value is calculated:

$$\chi^2_{n-1,\frac{\alpha}{2}} = \text{chi2inv}\left(1 - \frac{\alpha}{2}, n-1\right) = \text{chi2inv}\left(1 - \frac{0.01}{2}, 20-1\right) = 38.5823$$

And the lower critical value:

$$\chi^2_{n-1,1-\frac{\alpha}{2}} = \text{chi2inv}\left(\frac{\alpha}{2}, n-1\right) = \text{chi2inv}\left(\frac{0.01}{2}, 20-1\right) = 6.844$$

b. $\alpha = 0.05$ and $n = 12$.

$$\chi^2_{n-1,\frac{\alpha}{2}} = \text{chi2inv}\left(1 - \frac{\alpha}{2}, n-1\right) = \text{chi2inv}\left(1 - \frac{0.05}{2}, 12-1\right) = 21.92$$

$$\chi^2_{n-1,1-\frac{\alpha}{2}} = \text{chi2inv}\left(\frac{\alpha}{2}, n-1\right) = \text{chi2inv}\left(\frac{0.05}{2}, 12-1\right) = 3.8157$$

c. $\alpha = 0.10$ and $n = 15$.

$$\chi^2_{n-1,\frac{\alpha}{2}} = \text{chi2inv}\left(1 - \frac{\alpha}{2}, n-1\right) = \text{chi2inv}\left(1 - \frac{0.1}{2}, 15-1\right) = 23.6848$$

$$\chi^2_{n-1,1-\frac{\alpha}{2}} = \text{chi2inv}\left(\frac{\alpha}{2}, n-1\right) = \text{chi2inv}\left(\frac{0.1}{2}, 15-1\right) = 6.5706$$

10 Exercise 9.4.3

Consider a sample with $n = 17$ and sample std. dev. $s = 0.09$.

a. Is there strong evidence to conclude that the standard deviation exceeds 0.75? Use $\alpha = 0.05$. Find the P-value.

First we set up the hypothesis:

$$H_0 : \sigma^2 = 0.75^2, \qquad H_1 : \sigma^2 > 0.75^2$$

The test statistic is calculated using eq. 4:

$$x_0^2 = (n-1)\frac{s^2}{\sigma_0^2} = (17-1)\frac{0.09^2}{0.75^2} = 0.2304$$

This is a upper sided case so the critical value is calculated with MATLAB as:

$$\chi^2_{n-1,\alpha} = \text{chi2inv}(1-\alpha, n-1) = \text{chi2inv}(1-0.05, 17-1) = 26.3$$

Since $x_0^2 < \chi^2_{n-1,\alpha}$ we fail to reject H_0. The P-value is calculated as:

$$P = 1 - \text{chi2cdf}(x_0^2, n-1) = 1 - \text{chi2cdf}(0.2304, 17-1) = 1$$

b. Discuss how (a) could be answered with a one-sided confidence bound.

We want to test to see if the std. dev. is above 0.75. We can therefore make a one-sided lower bound CI on the std. dev.:

$$\left[\sqrt{\frac{(n-1)s^2}{\chi^2_{n-1,\alpha}}}, \infty\right[= \left[\sqrt{\frac{(17-1)\cdot 0.09^2}{26.3}}, \infty\right[= [0.07; \infty[$$

11 Exercise 9.5.2

Suppose that of 1000 customers surveyed, 850 were satisfied with a product.

a. Test the hypothesis $H_0 : p = 0.9$ against $H_1 : p \neq 0.09$ at $\alpha = 0.05$ and find the P-value.

This is a test on the population proportion. We can calulate the test statistic using eq. 5. here $X = 850$ and $n = 1000$:

$$z_0 = \frac{X - np_0}{\sqrt{np_0(1-p_0)}} = \frac{850 - 1000 \cdot 0.9}{\sqrt{1000 \cdot 0.9(1-0.9)}} = -5.2705$$

because this is a normal approximation of the binomial the lower test statistic is:

$$z_{\frac{\alpha}{2}} = \text{norminv}\left(\frac{\alpha}{2}\right) = \text{norminv}\left(\frac{0.05}{2}\right) = -1.96$$

Since $z_0 < z_{\frac{\alpha}{2}}$ it is in the critical region and we reject H_0. The P-value is for the two-sided test:

$$P = 2\left[1 - \text{normcdf}\left(|z_0|\right)\right] = 2\left[1 - \text{normcdf}\left(5.2705\right)\right] = 1.361 \cdot 10^{-7} \approx 0$$

b. Explain how to answer (a) with a two-sided confidence interval for p.

The two-sided CI for the population proportion is given as:

$$\left[\hat{p} \pm z_{\frac{\alpha}{2}}\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}\right]$$

Where:

$$\hat{p} = \frac{850}{1000} = 0.85$$

The CI is then:

$$\left[0.85 \pm 1.96\sqrt{\frac{0.85(1-0.85)}{1000}}\right] = [0.828; 0.872]$$